LAPORAN WORKSHOP DEVELOPER OPERATIONAL

MINGGU KE-11

OLEH:

AMELIA AGUSTIN

E41230432

GOLONGAN A

PROGRAM STUDI TEKNIK INFORMATIKA

JURUSAN TEKNOLOGI INFORMASI

POLITEKNIK NEGERI JEMBER

2026

HASIL DAN PEMBAHASAN

```
PS C:\WINDOWS\system32> docker --version
Docker version 29.2.1, build a5c7197
```

```
PS C:\WINDOWS\system32> docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED        STATUS        PORTS      NAMES
68d3c6b64b46   phpmyadmin/phpmyadmin   "/docker-entrypoint.…"   2 months ago   Up 1 second   80/tcp     pma
9707da97fa4e   mysql:5.7              "docker-entrypoint.s…"   2 months ago   Up 1 second   3306/tcp   mysql
```

```
PS C:\WINDOWS\system32> winget install Kubernetes.minikube
The `msstore` source requires that you view the following agreements before using.
Terms of Transaction: https://aka.ms/microsoft-store-terms-of-transaction
The source requires the current machine's 2-letter geographic region to be sent to the backend service to function prope
rly (ex. "US").
```

```
Do you agree to all the source agreements terms?
[Y] Yes  [N] No: yes
Found Kubernetes - Minikube - A Local Kubernetes Development Environment [Kubernetes.minikube] Version 1.38.1
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.
Downloading https://github.com/kubernetes/minikube/releases/download/v1.38.1/minikube-installer.exe
  ▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒▒  48.5 MB / 48.5 MB
Successfully verified installer hash
Starting package install...
Successfully installed
```

```
PS C:\WINDOWS\system32> minikube version
minikube version: v1.38.1
commit: c93a4cb9311efc66b90d33ea03f75f2c4120e9b0
```

```
PS C:\WINDOWS\system32> minikube start --driver=docker
* minikube v1.38.1 on Microsoft Windows 11 Pro 25H2
* Using the docker driver based on user configuration
! Starting v1.39.0, minikube will default to "containerd" container runtime. See #21973 for more info
* Using Docker Desktop driver with root privileges
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.50 ...
* Downloading Kubernetes v1.35.1 preload ...
    > preloaded-images-k8s-v18-v1...:  272.45 MiB / 272.45 MiB  100.00% 3.62 Mi
    > gcr.io/k8s-minikube/kicbase...:  519.58 MiB / 519.58 MiB  100.00% 2.61 Mi
* Creating docker container (CPUs=2, Memory=3072MB) ...
* Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
* Configuring bridge CNI (Container Networking Interface) ...
* Verifying Kubernetes components...
  - Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Enabled addons: storage-provisioner, default-storageclass
* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
```

```
PS C:\WINDOWS\system32> kubectl get nodes
NAME       STATUS   ROLES           AGE     VERSION
minikube   Ready    control-plane   5m46s   v1.35.1
```

```
PS C:\Users\user> mkdir kubernetes


    Directory: C:\Users\user


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----         4/29/2026   8:37 AM                kubernetes
```

```
PS C:\Users\user> cd kubernetes
PS C:\Users\user\kubernetes> code .
```

EXPLORER

KUBERNETES

- deployment.yaml
- service.yaml

Welcome | deployment.yaml | service.yaml

deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: web-nginx
spec:
  replicas: 3
  selector:
    matchLabels:
      app: web-nginx
  template:
    metadata:
      labels:
        app: web-nginx
    spec:
      containers:
      - name: nginx
        image: nginx
        ports:
        - containerPort: 80
```

EXPLORER

KUBERNETES

- deployment.yaml
- service.yaml

Welcome | deployment.yaml | service.yaml

service.yaml

```yaml
apiVersion: v1
kind: Service
metadata:
  name: web-nginx-service
spec:
  selector:
    app: web-nginx
  ports:
    - port: 80
      targetPort: 80
  type: NodePort
```

```
PS C:\Users\user\kubernetes> kubectl apply -f deployment.yaml
deployment.apps/web-nginx created
PS C:\Users\user\kubernetes> kubectl apply -f service.yaml
service/web-nginx-service created
```

```
PS C:\Users\user\kubernetes> kubectl get pods
NAME                          READY   STATUS    RESTARTS   AGE
web-nginx-748d764f7-25cqz     1/1     Running   0          2m5s
web-nginx-748d764f7-dwgg8     1/1     Running   0          2m5s
web-nginx-748d764f7-sv5t2     1/1     Running   0          2m5s
```

```
PS C:\Users\user\kubernetes> kubectl get all
NAME                                READY   STATUS    RESTARTS   AGE
pod/web-nginx-748d764f7-25cqz       1/1     Running   0          2m53s
pod/web-nginx-748d764f7-dwgg8       1/1     Running   0          2m53s
pod/web-nginx-748d764f7-sv5t2       1/1     Running   0          2m53s

NAME                          TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
service/kubernetes            ClusterIP   10.96.0.1       <none>        443/TCP        23m
service/web-nginx-service     NodePort    10.99.169.224   <none>        80:31773/TCP   2m4s

NAME                          READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/web-nginx     3/3     3            3           2m53s

NAME                                     DESIRED   CURRENT   READY   AGE
replicaset.apps/web-nginx-748d764f7      3         3         3       2m53s
```

```
PS C:\Users\user\kubernetes> kubectl get pods -w
NAME                          READY   STATUS    RESTARTS   AGE
web-nginx-748d764f7-25cqz     1/1     Running   0          3m22s
web-nginx-748d764f7-dwgg8     1/1     Running   0          3m22s
web-nginx-748d764f7-sv5t2     1/1     Running   0          3m22s
```

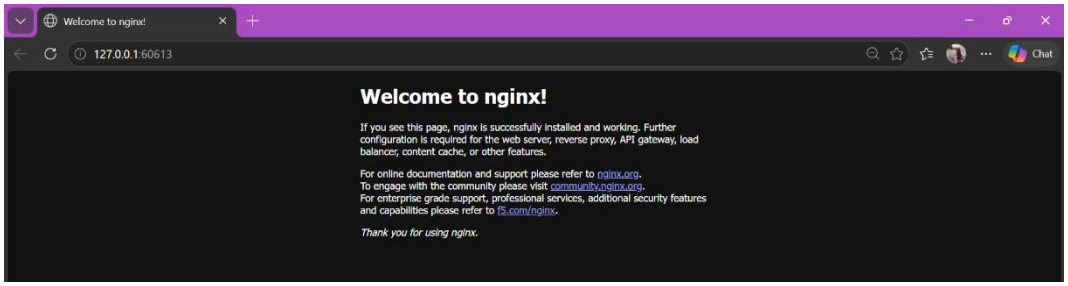
```
PS C:\Users\user\kubernetes> minikube service web-nginx-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	web-nginx-service	80	http://192.168.49.2:31773

```
🔗  Starting tunnel for service web-nginx-service.
```

NAMESPACE	NAME	TARGET PORT	URL
default	web-nginx-service		http://127.0.0.1:60613

```
🎉  Opening service default/web-nginx-service in default browser...
❗  Because you are using a Docker driver on windows, the terminal needs to be open to run it.
```



PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTMAN CONSOLE

```
PS C:\Users\user\kubernetes> kubectl get pods -w
NAME                          READY   STATUS    RESTARTS   AGE
web-nginx-748d764f7-25cqz     1/1     Running   0          7m44s
web-nginx-748d764f7-dwgg8     1/1     Running   0          7m44s
web-nginx-748d764f7-sv5t2     1/1     Running   0          7m44s
web-nginx-748d764f7-j2pql     0/1     Pending   0          0s
web-nginx-748d764f7-s88gn     0/1     Pending   0          1s
web-nginx-748d764f7-j2pql     0/1     Pending   0          1s
web-nginx-748d764f7-s88gn     0/1     Pending   0          1s
web-nginx-748d764f7-j2pql     0/1     ContainerCreating   0          1s
web-nginx-748d764f7-s88gn     0/1     ContainerCreating   0          1s
web-nginx-748d764f7-s88gn     1/1     Running             0          11s
web-nginx-748d764f7-j2pql     1/1     Running             0          13s
web-nginx-748d764f7-j2pql     1/1     Terminating         0          20s
web-nginx-748d764f7-s88gn     1/1     Terminating         0          20s
web-nginx-748d764f7-j2pql     1/1     Terminating         0          20s
web-nginx-748d764f7-s88gn     1/1     Terminating         0          21s
web-nginx-748d764f7-s88gn     0/1     Completed           0          25s
web-nginx-748d764f7-j2pql     0/1     Completed           0          25s
web-nginx-748d764f7-j2pql     0/1     Completed           0          26s
web-nginx-748d764f7-j2pql     0/1     Completed           0          26s
web-nginx-748d764f7-s88gn     0/1     Completed           0          26s
web-nginx-748d764f7-s88gn     0/1     Completed           0          26s
web-nginx-748d764f7-s88gn     0/1     Completed           0          26s
```

kubectl
minikube
terminal 1
terminal 2
terminal 3

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTMAN CONSOLE

```
PS C:\Users\user\kubernetes> kubectl scale deployment web-nginx --replicas=5
deployment.apps/web-nginx scaled
PS C:\Users\user\kubernetes> kubectl scale deployment web-nginx --replicas=3
deployment.apps/web-nginx scaled
PS C:\Users\user\kubernetes>
```

kubectl
minikube
terminal 1
terminal 2
terminal 3

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS POSTMAN CONSOLE

```
PS C:\Users\user\kubernetes> kubectl get deployment -w
NAME        READY   UP-TO-DATE   AVAILABLE   AGE
web-nginx   3/3     3            3           8m42s
```

kubectl
minikube
terminal 1
terminal 2
terminal 3

```
PS C:\Users\user\kubernetes> minikube node add
😄  Adding node m02 to cluster minikube as [worker]
❗  Cluster was created without any CNI, adding a node to it might cause broken networking.
👍  Starting "minikube-m02" worker node in "minikube" cluster
🚜  Pulling base image v0.0.50 ...
```

```
PS C:\Users\user\kubernetes> kubectl get nodes
NAME       STATUS   ROLES           AGE   VERSION
minikube   Ready    control-plane   32m   v1.35.1
```

```
PS C:\Users\user\kubernetes> minikube status
E0429 08:57:32.021088    1320 status.go:183] the "minikube-m02" host does not exist
E0429 08:57:32.022414    1320 status.go:127] status error: the "minikube-m02" host does not exist
```

```
PS C:\Users\user\kubernetes> kubectl delete -f deployment.yaml
deployment.apps "web-nginx" deleted from default namespace
PS C:\Users\user\kubernetes> kubectl delete -f service.yaml
service "web-nginx-service" deleted from default namespace
```